them. Finally, we will look at the two types of injection attack: SQL injection, a vulnerability that can result in arbitrary SQL commands being executed against your database; and cross-site scripting (XSS), where a malicious user can cause havoc and steal data by injecting client-side scripts into your pages.

Many threats can be mitigated simply by implementing HTTPS. In 2014, Google announced it would include HTTPS as a ranking signal (http://mng.bz/aP4m). The intention behind this announcement was to persuade as many site owners as possible to improve site security by running their application over encrypted connections backed up by valid certificates. To begin our look at application security, you will learn about the importance of HTTPS and learn how you can configure your application to work with it.

13.1 HTTPS

I'll start with a review of the basics. What is HTTPS, and why would you want to use it? HTTPS is a protocol that ensures a secure connection between your website and your users. HTTPS provides security in three ways:

- *Authentication*—Users can be sure they are interacting with your site rather than a spoofed copy hosted under a different domain.
- *Encryption*—All communication between the user and your website is encrypted, ensuring it cannot be stolen.
- *Data integrity*—Users are guaranteed the content they receive has not been modified by a third party.

Communication over an unsecured connection (HTTP) is conducted in plain text. Anyone who has access to the network you are using can see the content of your communications, including posted form values, such as your username and password, for example. They can also insert their own content into the response before it reaches you. Some of us are old enough to remember the *free WiFi* services offered by hotels that would insert their own advertising or branding into the web page you were trying to reach. Now imagine someone with less clear motives inserting their own links into pages you felt you could trust, and you will begin to understand the potential harm that can come from interacting with unsecured websites.